LastWell

Stroke alerts that start on the 911 call



EMS phase lengths are to scale, from national chest-pain medians (source 9). Stroke timings may differ.

A voice agent that listens to a 911 medical call, recognizes a likely stroke from what the caller says and how they sound, and pins down the one fact that decides treatment: when the patient was last seen normal. With one click from the dispatcher, it briefs the hospital stroke team by voice before the ambulance has even arrived.

Target tracks	Anthropic (Claude) track, Microsoft Grand Prize: Built for the Real World, Best Use of Cartesia, general prizes: Best People-First Hack, Most Technically Complex
What judges see	A judge calls 911 from their own phone, describes a sick relative, and 40 seconds later their phone rings again: a stroke-team voice briefing they can question out loud.
Stack	Twilio voice, Cartesia Ink 2 (streaming speech-to-text) and Sonic 3.5 (speech), Claude Haiku 5.5 and Sonnet 5.5, Next.js, Supabase Realtime
Status	Concept and build plan. Nothing in this brief has been built yet; every result marked [TBD] gets filled in during the event.

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How to use this brief. Sections 1 to 3 are the argument; read them before pitching. Sections 4 to 8 are what you build and show. Sections 9 to 13 are for the weekend itself: track criteria, the hour-by-hour plan, judge questions and a Devpost draft.

1 At a glance

> **The one-sentence pitch.** Today a hospital usually hears about an incoming stroke from paramedics who are already standing next to the patient; LastWell starts that warning from the first minute of the 911 call and delivers the two facts that decide treatment.

	Detail
Who it serves	Older adults having a stroke at home, where a spouse, a child or the patient calls 911 and often never says the word "stroke". The users are the 911 dispatcher and the on-call stroke neurologist.
The gap	Dispatchers miss many strokes (about half in a large Copenhagen study [1]). Paramedic pre-alerts reached the hospital for only 67% of EMS-transported stroke patients in a US registry of 371,988 patients [6]. In a survey of EMS providers, 39.5% named information lost in dispatch as a barrier to pre-alerting [7].
What LastWell does	Listens to the call, flags a likely stroke with quotes from the call as evidence, prompts the dispatcher to ask when the patient was last seen normal, and sends a dispatcher-approved pre-alert to the stroke team by text and by an interactive voice briefing.
What is new	Stroke detection on emergency calls exists only as European research [1][4][5]. US products cover general transcription and triage [11][12], and hospital pre-alerts start from paramedics [10]. Nobody we found sends a stroke pre-alert from the call itself.
Why judges believe it	The judge is the caller. Their own words drive the flag, and their own phone receives the briefing. We also show a scored test set against a keyword baseline.
Honest limits	No real 911 audio (we use scripted and acted calls). Detection is imperfect: even the research model was right only about a quarter of the time it flagged a stroke [1]. The dispatcher stays in charge of every alert.

2 The problem

A stroke starts a clock. Clot-dissolving IV tPA has to be given within 4.5 hours of the time the patient was **last known well**, and earlier treatment goes with better outcomes [7]. Three things slow the race down before the patient reaches a CT scanner.

1. The 911 call often doesn't sound like a stroke

Callers describe confusion, a fall, dizziness, a drunk-sounding husband. In a Copenhagen study of more than 1.5 million emergency calls, human call handlers recognized 52.7% of strokes; an AI model trained on the same calls recognized 63.0% [1]. Strokes are rare in call volume: in that data, about 4 in 1,000 calls were strokes [2], which makes them easy to miss on a busy shift. Norwegian researchers are now studying the strokes call handlers miss, which they call "stroke chameleons" [5].

2. The hospital hears late, or not at all

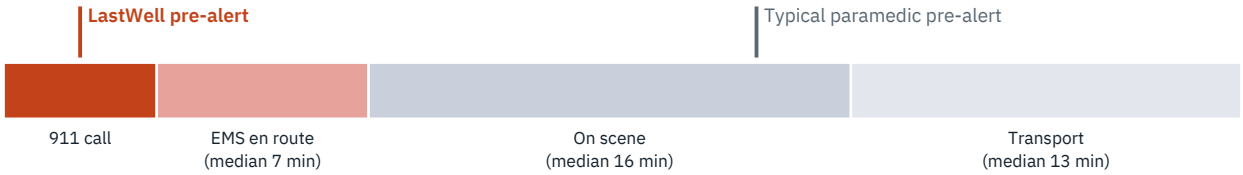
When paramedics pre-alert the hospital, stroke care is faster. In the Get With The Guidelines-Stroke registry, pre-notification was linked to faster imaging, faster door-to-needle times and more eligible patients getting tPA, yet it happened for only 67.0% of EMS-transported patients [6]. A single-center registry study found shorter door-to-CT times when EMS documented a last-known-well time and when they pre-notified the hospital [8].

3. Details get lost between the call and the ambulance

In a survey of 301 EMS providers, 39.5% said information lost in dispatch was a barrier to pre-notifying the hospital, and only 11.0% correctly named the 4.5-hour tPA window measured from last known well [7]. The caller often knows exactly when Dad was last normal. That fact needs to survive the hand-offs.

How many minutes are on the table

We have not found US stroke-specific numbers yet, so we use national EMS data for chest pain as a stand-in: median response time 7 minutes, median on-scene time 16 minutes, median transport time 13 minutes [9]. A paramedic pre-alert normally goes out after the on-scene assessment. A pre-alert sent during the call could therefore reach the stroke team roughly **7 to 23 minutes earlier** in a median call. That is an estimate to verify, and the pitch should call it one.



EMS phases are to scale, from the chest-pain medians in [9]; the call segment is illustrative. Stroke timings may differ.

The named community (Anthropic track wording)

> **Older adults in San Francisco having a stroke at home, whose spouse or adult child calls 911 and describes it as confusion, a fall or "he sounds drunk"**, plus the dispatchers and on-call stroke neurologists responsible for them.
>
> Before Saturday noon, at least one teammate talks to someone who has lived this: an EMT, nurse, dispatcher or a family member of a stroke patient (mentors and Slack count). Their quote goes in the Devpost: [quote and role].

3 What already exists, and the gap

911 voice AI is a crowded field, which is why LastWell stays narrow. The table shows who owns what.

Product or project	What it does	What it leaves open
Carbyne (now part of Axon)	Live transcription and translation, call triage and real-time call summaries; it also sends video and location to responders [11].	General purpose. No stroke-specific logic or hospital pre-alert found.
Prepared (now part of Axon)	Transcripts, translation, keyword alerts to supervisors, for example when someone is not breathing [12].	Alerts stay inside the 911 center.
Corti	Reviews every emergency call against dispatch standards; trained on 13 million+ calls [13]. A separate Danish system that detects cardiac arrest on calls is in use across Denmark [3].	Call review against standards; no stroke pre-alert found.
Knogin audio intelligence	Detects gunshots, alarms, agonal breathing and caller stress during calls [14].	Sound events, not stroke reasoning.
Motorola Assist	Voice assistant for responders en route; shares live 911 audio with field units [15].	Field-facing; nothing aimed at the hospital.
Pulsara	EMS-to-hospital alerts with ETA, chat, images, ECG [10].	Starts once paramedics are with the patient.
DispatchAI (UC Berkeley AI Hackathon 2024 grand prize)	Dispatcher copilot: live transcript, severity map, AI summary, human in the loop [16].	General triage. Judges may remember it, so we must not look like it.
Copenhagen and Bergen research	Machine learning that flags stroke on emergency calls [1][4][5].	Not a product in the US. No hospital hand-off. Speech features such as slurring are listed as future work [1].

> **LastWell's lane:** the first usable version of a research result, aimed at a hand-off nobody owns. The call-to-stroke-team link starts minutes before Pulsara's paramedic alert, and the stroke reasoning goes deeper than the keyword alerts in general 911 tools.

Pitch order follows from this table. Open with stroke ("half of strokes are missed on the call"), never with "AI for 911", so judges don't file LastWell next to DispatchAI.

4 The product

LastWell has two screens and one phone call. Everything else stays behind the scenes.

Screen 1: the dispatcher console

- **Live transcript** of both sides of the call, with words that point to stroke underlined as they appear.
- **Stroke likelihood card** with a level (none yet / possible / likely), the BE-FAST signs heard so far (balance, eyes, face, arms, speech, time) and the exact caller quote behind each one. Clicking a quote jumps to that moment in the transcript.
- **Next question** prompt, one at a time, for example "Ask: when was he last seen completely normal?" or "Ask: does he take blood thinners?"
- **Look-alike check** that shows what else could explain the call (low blood sugar, intoxication, seizure) and the question that would separate them, such as "Is he diabetic?"
- **Critical facts** collected as the call goes: last known well as a clock time, blood thinners, age, diabetes, address.
- **One button: Send stroke pre-alert.** Nothing leaves the 911 center until the dispatcher presses it.

The phone call: the stroke-team voice briefing

When the dispatcher approves, the on-call stroke neurologist's phone rings. A Cartesia Sonic voice gives a 20-second briefing: "Stroke pre-alert from 911. 78-year-old man, right-sided weakness and slurred speech per his wife. Last seen normal at 2:10 PM, 47 minutes ago. Wife says he takes Eliquis. Ambulance dispatched." The neurologist can then ask questions out loud ("Any seizure before this?") and LastWell answers only from the call, with "That wasn't mentioned on the call" when it doesn't know. A text message carries the same facts.

Screen 2: the stroke-team board

- One card per incoming pre-alert, showing the facts with their quotes and the time since last known well.
- A countdown to the 4.5-hour mark from last known well [7], with the caveat that the treating team makes every treatment decision.
- Status that upgrades when paramedics confirm on scene, so the hospital can tell a dispatcher pre-alert from a paramedic-confirmed one.

What LastWell will never do

> Dispatch units, change call priority, give medical instructions to the caller, or tell anyone "this is not a stroke". When it hears nothing, the card says "No stroke signs heard yet". The dispatcher's normal protocol runs exactly as before.

5 How it works

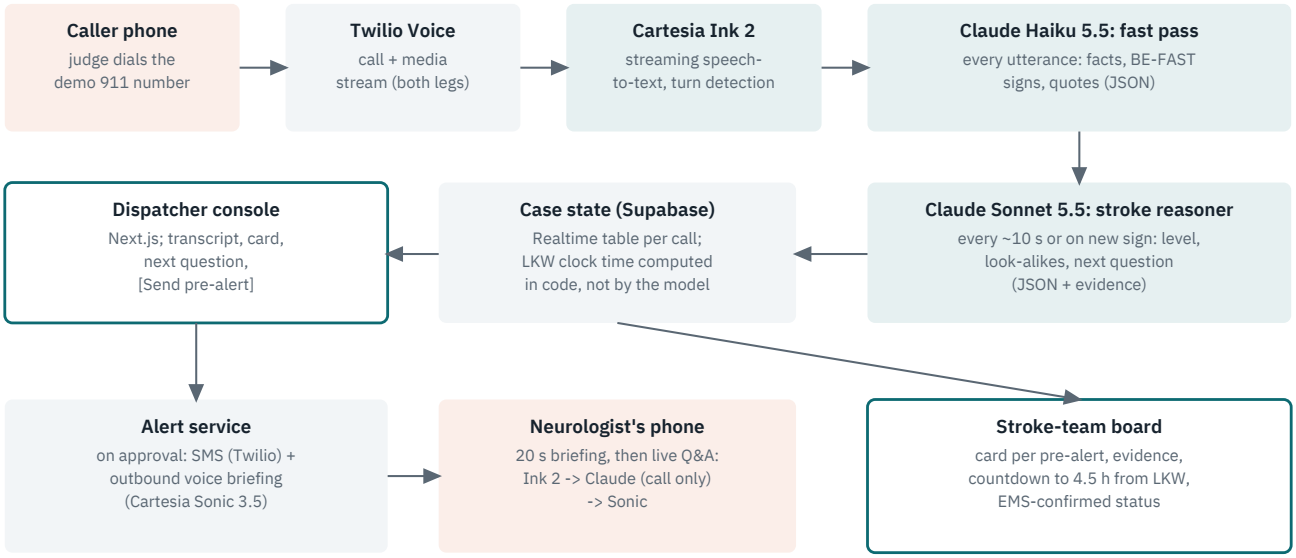


Figure 1. Data flow. Tinted teal boxes are model calls, outlined boxes are the two screens people use, and warm boxes are phones.

Two speeds of reasoning

A small model is meant to read each utterance within about a second and pull out facts and quotes. A larger model re-reads the whole transcript every ten seconds, or as soon as a new sign appears, and decides the likelihood level, the look-alikes and the next question. The fast pass keeps the console live; the slow pass keeps the judgment careful. The demo shows the latency of each in a small corner readout [TBD after build].

Output contract

Both passes must return JSON that the console validates. Every sign and fact carries the quote and timestamp it came from; anything without a quote is dropped. Example from the Sonnet pass:

```
{
  "level": "likely",
  "signs": [
    {"sign": "speech", "quote": "his words are all mixed up", "t": "00:41"},
    {"sign": "arms", "quote": "he can't lift his coffee cup", "t": "00:58"}
  ],
  "last_known_well": {"phrase": "he was fine at lunch, around two", "t": "01:12"},
  "blood_thinner": {"value": "Eliquis", "quote": "he's on Eliquis", "t": "01:30"},
  "lookalikes": [{"name": "low blood sugar", "ask": "Is he diabetic?"}],
  "next_question": "Ask whether he has diabetes."
}
```

Time is computed by code

Claude only extracts the phrase ("around two", "about an hour ago"). A small parser turns it into a clock time using the call's start time, keeps a range when the phrase is vague, and shows the original words beside the result. Clock arithmetic is exactly the kind of thing a language model can get wrong, and a wrong last-known-well time could change a treatment decision.

6 Where Claude could go wrong, and what we do about it

This section is written for the Anthropic track's fourth criterion and Microsoft's "responsible design" criterion. Each risk has a design answer you can point to on screen.

Failure	Harm	What LastWell does
False alarm	Stroke teams start ignoring pre-alerts (alert fatigue).	Dispatcher approval for every alert; "pre-alert" level kept separate from paramedic-confirmed; precision reported honestly in the eval.
Missed stroke	False reassurance.	The card never says "not a stroke", only "no stroke signs heard yet". Normal dispatch protocol is untouched.
Invented facts	A made-up medication or time reaches the hospital.	Every fact needs a quote and timestamp from the transcript or it is dropped; the voice briefing answers "not mentioned" when unknown.
Wrong last-known-well time	Could push a patient in or out of a treatment window on paper.	Claude extracts the phrase; code computes the time; the original words are always shown; vague phrases stay ranges.
Accent or panic bias	Worse detection for some callers.	The Danish team reported their system struggled with foreign accents and panicked speech [3]. Our test set includes non-native speakers and panicked callers, and we report results by group [TBD].
Automation bias	Dispatcher defers to the screen.	Prompts are questions to ask, never conclusions; the console has no "dispatch" button.
Privacy	Health details of a caller in crisis.	Demo uses scripted calls only. Audio is not stored after the call in the prototype; production would need the security reviews 911 vendors go through.

Why Claude for this job

The hard part is judgment over messy, partial speech: noticing that "he sounds drunk but he doesn't drink" is a stroke sign, weighing low blood sugar as a look-alike, and refusing to state anything the caller never said. The output contract and quote requirement let us check that judgment on every call. We do not have to take it on trust.

7 Proving it works

Real 911 recordings with stroke outcomes are not public, so we build a scripted test set and say so on the slide. The goal is an honest comparison against a baseline, run on Saturday evening.

Call type	Count	Example
Classic stroke	10	Wife reports face drooping and slurred speech after lunch.
Stroke chameleon	8	"He's dizzy and keeps falling, probably just tired"; "she's confused, maybe drunk".
Look-alike, not stroke	8	Diabetic with low blood sugar; seizure; heavy drinking; migraine.
Other medical	6	Chest pain, fall with hip pain, breathing trouble.
Hard audio	6	Non-native speaker, panicked caller, TV in the background (re-reads of earlier scripts).

Scripts are written from published stroke presentations and reviewed by whoever on the team has clinical exposure, or a mentor. Teammates record them as audio so the full pipeline (speech-to-text included) is tested.

What we measure

Metric	Why	LastWell	Keyword baseline
Sensitivity (strokes flagged)	Missed strokes are the main harm	[TBD]	[TBD]
Precision (flags that were strokes)	Alert fatigue	[TBD]	[TBD]
Seconds into call at first flag	Earlier is the point	[TBD]	[TBD]
Last-known-well extracted correctly	Decides treatment	[TBD]	n/a
Blood thinner extracted correctly	Can rule out tPA	[TBD]	n/a
Invented facts	Must be zero	[TBD]	n/a

The keyword baseline flags any call containing words like "stroke", "droop", "slurred" or "numb". If LastWell doesn't beat it on chameleons, we say so and show the cases it missed. The Copenhagen result (63.0% recall, 24.9% precision for the AI versus 52.7% and 17.1% for call handlers [1]) is the reference point for what a good real-world number looks like; our scripted numbers will look better than reality and the slide should say that.

8 The demo

Judges see a live demo for about two minutes at a shared table [18]. LastWell's demo is built so the judge does the important parts themselves.

Table setup

- Laptop 1: dispatcher console, facing the judge.
- Laptop 2 or a tablet: stroke-team board, at the other end of the table.
- Three scenario cards (below) and a printed QR or phone number for the demo line.
- A teammate wearing a headset plays the dispatcher and answers the judge's call.
- Backup: a 90-second screen recording of the full flow, cued up, in case the venue network fails.

Two-minute script

Time	What happens	Who
0:00	"In a big Copenhagen study, call handlers caught about half of strokes on the 911 call [1]. And the hospital usually hears only once paramedics are with the patient."	Presenter
0:15	Hand the judge Card A. "Call this number and tell our dispatcher what's happening."	Presenter
0:20	Judge describes Dad. Console underlines "words are all mixed up", the card moves to Possible, then Likely; the prompt says "Ask when he was last seen normal."	Judge + dispatcher
0:50	Dispatcher asks; judge answers "around two". Card shows 2:00 PM (range 1:45 to 2:15) beside the judge's words. Dispatcher clicks Send stroke pre-alert.	Dispatcher
1:00	**The judge's own phone rings**: the stroke-team briefing in a Sonic voice. Invite them to ask it something ("Is he on blood thinners?").	Judge
1:25	Point at the board: card, quotes, countdown. Then 15 seconds on Card B (chameleon, flagged) and Card C (low blood sugar, held back with "Is he diabetic?").	Presenter
1:45	One slide: test-set results versus keyword baseline, labeled as scripted calls. Close on what's next: shadow-mode pilot with a 911 center.	Presenter

Scenario cards

Card	Judge reads	Expected result
A: classic	You're calling about your father, 78. Since lunch his words come out wrong and his right hand is weak. He was fine at about 2 PM. He takes Eliquis.	Likely stroke, LKW about 2 PM, blood thinner noted, pre-alert sent.
B: chameleon	Your wife is dizzy, keeps bumping into the wall and says she sees double. You think she's just tired.	Possible stroke (balance, eyes), prompts for face and arm checks.
C: look-alike	Your husband is diabetic, skipped dinner, is sweaty and confused.	Look-alike flagged (low blood sugar); LastWell asks about diabetes and holds the pre-alert.

> **Demo rules.** Card A is the only live call that must work; B and C can run from recorded audio. Rehearse the full script ten times on Sunday morning with the real phone line. If the phone line fails, switch to the browser microphone (same pipeline) without apologizing. The Cal Hacks 12.0 winners hardcoded a fallback for the one part of their agent that was unreliable and cut it from the demo [17].

9 Track-by-track fit

Criteria below are quoted or paraphrased from the Cal Hacks 13.0 Hacker Guide [18]. Prize details marked "coming soon" in the guide should be re-checked on the live site Friday.

Anthropic (prize: coming soon)

Criterion	LastWell's answer
A named community	Older adults in SF having a stroke at home, whose relative calls 911 (Section 2).
A real problem you lived or heard about	Interview with an EMT, nurse, dispatcher or family member before Saturday noon; quote in Devpost.
It works	One flow done well: call, flag, pre-alert, briefing. Usable as a shadow-mode tool on Monday.
Thoughtful use of Claude	Two-speed reasoning, quote-or-drop facts, code-computed time, explicit failure table (Section 6).

Microsoft Grand Prize: Built for the Real World (Surface laptop per member)

Criterion	LastWell's answer
Customer understanding	Two users with different needs: dispatcher (one prompt at a time) and neurologist (facts in 20 seconds).
Measurable usefulness	Sensitivity, precision, seconds to flag, extraction accuracy versus baseline (Section 7).
Path from prototype to production	Shadow mode, then pre-alert pilot with one 911 center and one stroke center (Section 12).
Responsible design	Human approval on every alert, no autonomous dispatch, bias testing (Section 6).
End-to-end functionality	Real phone call in, real phone call out, live boards in between.

Best Use of Cartesia (prize listed as Meta glasses, TBD)

Criterion (25% each)	LastWell's answer
Problem and originality	A clinical hand-off nobody owns, scoped to one condition.
Use of Cartesia	Voice is the product on both ends: Ink 2 streaming speech-to-text listens to the call; Sonic 3.5 speaks the briefing; Ink 2 + Sonic run the neurologist's live Q&A; [19].
Execution	Main flow works live; edge cases are the chameleon and look-alike cards.
Impact and responsibility	Accessibility of a hands-free briefing for a neurologist who may be driving in; privacy and bias handling in Section 6.
Submission extras they ask for	2-minute demo-first presentation, README with Cartesia features used and known limits, no keys in the repo [18].

General prizes

Best People-First Hack fits the community framing; Most Technically Complex fits a real-time two-call voice pipeline with two model speeds. First to third overall are judged on impact, quality, technical complexity and creativity [18].

10 Build plan for 36 hours

All code is written at the event (the guide disqualifies projects started earlier [18]). Planning documents like this one are fine.

Roles for a team of four

Role	Owns
Voice	Twilio number and media stream, Cartesia Ink 2 streaming, outbound briefing call with Sonic, Q&A; loop.
Reasoning	Claude prompts and JSON validation, the last-known-well parser, the test runner with its keyword baseline.
Screens	Dispatcher console, stroke-team board, Supabase Realtime, the demo's look.
Clinical and pitch	Interview, scenario cards, 38 test scripts and recordings, Devpost, slide, rehearsal lead.

On a team of three, the other roles share Clinical and pitch work; with five, add a second person on Voice.

Timeline with go / no-go gates

When	Milestone	If it fails
Fri 6 to 9 PM	Repos, keys, Twilio number. Spike: phone call audio reaches Cartesia and prints a transcript.	Keep trying until gate 1.
Fri 11 PM (gate 1)	Live transcript from a real phone call on screen.	Switch the demo to browser microphone audio; keep phone for the briefing only.
Sat 4 AM (gate 2)	Haiku pass returns valid JSON with quotes on 5 test scripts.	Simplify the schema; drop look-alikes to v2.
Sat 10 AM	Dispatcher console live with card, quotes, next question. Interview done by noon.	-
Sat 2 PM (gate 3)	Approve button sends SMS and places the Sonic briefing call.	One-way briefing without Q&A.;
Sat 6 PM	Stroke-team board; Sonnet pass; LKW parser.	Board can be cut; SMS + call carry the demo.
Sat 9 PM	Test runner scores all scripts; numbers go on the slide.	Run on a subset; say so.
Sat 11 PM	Devpost draft submitted (guide recommends a Saturday draft [18]); backup video recorded.	-
Sun 2 AM	Feature freeze. Only bug fixes after this.	-
Sun 6 to 8 AM	Ten full rehearsals on the real line; final Devpost edits.	-
Sun 9 AM	Submission deadline. Aim for 8:00.	-
Sun 10:30 AM	Whole team at the table for the full judging window [18].	-

Cut list, in order

If the clock runs out, cut in this order and keep the demo flow intact: (1) slurred-speech audio cue, (2) Q&A; on the briefing call, (3) stroke-team board, (4) the real phone line for the caller. The thing never to cut is the judge's phone ringing with the briefing.

11 Risks and judge questions

Risk	Likelihood	Plan
Phone audio latency or quality in a loud hall	Medium	Test in the venue Friday night; headset for the dispatcher; browser fallback.
Model misreads the judge's improvised call	Medium	Scenario cards keep calls on script; Card A tested many times; quotes on screen show what it heard.
Twilio or Cartesia account limits	Low to medium	Set up and top up accounts Friday; ask sponsor booths for credits.
Looks like the Berkeley DispatchAI winner	Medium	Stroke-first pitch, hospital briefing as the climax, no general triage features.
A clinician judge doubts the time savings	Medium	Show the timeline as an estimate from EMS data [9] and lean on the second benefit: last-known-well and blood thinner arrive intact [7][8].

Likely judge questions

Isn't this what Carbyne or Prepared do?

> They transcribe and triage every call type inside the 911 center [11][12]. LastWell does one condition deeply and reaches outside the center to the stroke team, which they don't do.

How many false alarms?

> Our scripted precision is [TBD]. Real-world precision will be lower; the research model was right about a quarter of the time it flagged [1]. That is why a human approves every alert and why pre-alerts are labeled as such.

Where is your data from?

> Scripted calls based on published stroke presentations, recorded by us. No real 911 audio, and we say so.

What if it misses a stroke?

> Then the call goes exactly as it does today. LastWell never tells anyone a call is not a stroke.

Would a hospital act on a dispatcher's pre-alert?

> Some hospitals may only acknowledge it until paramedics confirm. Even then, the last-known-well time and blood thinner status arrive early and intact, and EMS documentation of last known well is linked with faster CT [8].

How much time does it save?

> In a median call, roughly 7 to 23 minutes earlier than a pre-alert sent from the scene, based on national EMS medians for chest pain [9]. Stroke-specific numbers are on our list to verify.

What about accents?

> The Danish team reported exactly this problem [3]. Our test set includes non-native and panicked speakers, reported separately.

12 Path to production

This is the Microsoft track's fourth criterion. The plan moves from zero risk to real alerts in three stages.

Stage	What happens	Exit criterion
1. Retrospective	Run LastWell on a 911 center's archived medical calls, matched to hospital stroke outcomes (the approach the Copenhagen and Bergen teams used [1][5]).	Sensitivity and precision at least match call handlers.
2. Shadow mode	Runs live on calls; dispatchers see nothing; flags are compared with what happened.	Stable precision; no invented facts; agreement on last known well.
3. Pre-alert pilot	One 911 center and one stroke center; dispatcher-approved pre-alerts reach the stroke team.	Door-to-CT and door-to-needle times compared with the pre-pilot baseline.

Integration notes

- 911 centers run computer-aided dispatch systems; LastWell would attach its findings to the call record. Industry vendors already append acoustic findings to calls using the IETF RFC 7852 additional-data standard [14], a model for how LastWell could plug in.
- Hospitals that use Pulsara already receive paramedic alerts there [10]; sending LastWell pre-alerts into the same channel avoids one more app.
- Clinical decision support in emergency settings needs regulatory and legal review, and 911 data needs the security review that 911 vendors go through. Both are named as open questions in the pitch.

What's next after the hackathon

- Speech features (slurred speech, word-finding pauses) as a second signal; the Danish team listed this as future work [1].
- Routing to the right kind of stroke center when large-vessel stroke signs are reported.
- Spanish and Cantonese calls, both common in San Francisco, using Cartesia's multilingual models [19].

13 Devpost draft

Brackets are for things only the team can fill in. Edit in your own words before submitting.

Inspiration

[One line from the person you interviewed, with their role.] Stroke treatment depends on the time the patient was last seen normal, and the person who knows that time is usually the one on the 911 call. In a large Copenhagen study, call handlers recognized about half of strokes [1], and US registry data shows hospitals got a paramedic pre-alert for two thirds of EMS-transported stroke patients [6]. We wanted the stroke team to hear about it from the first minute of the call.

What it does

LastWell listens to a 911 medical call, flags a likely stroke with quotes from the call as evidence, prompts the dispatcher to ask when the patient was last seen normal and whether they take blood thinners, and, after one click from the dispatcher, calls the on-call stroke neurologist with a 20-second voice briefing they can question. A board at the hospital shows the case and the time since last known well.

How we built it

Twilio streams both sides of the call to Cartesia Ink 2 for live transcription. Claude Haiku 5.5 extracts facts and quotes from each utterance; Claude Sonnet 5.5 re-reads the transcript to set the stroke level, check look-alikes and pick the next question. Every fact must carry a quote and timestamp. A small parser turns "around two" into a clock time; the model never does that math. The briefing call uses Cartesia Sonic 3.5, and the boards are Next.js on Supabase Realtime. [Adjust to what you actually built.]

Challenges we ran into

[Fill in during the event: latency, phone audio, a misread call.]

Accomplishments that we're proud of

[Test-set result versus baseline, the first time the briefing call rang, anything a mentor said.]

What we learned

[Fill in honestly.]

What's next for LastWell

Shadow-mode testing on archived calls with a 911 center, speech features for slurred speech, and Spanish and Cantonese support.

Built with

claude-api, cartesia, twilio, next.js, supabase, typescript, python

Submission checklist

- Every teammate added on Devpost; draft live by Saturday night [18].
- Repo public for judges, README with setup, Cartesia features used, known limitations; no API keys [18].
- Sponsor tracks selected: Anthropic, Microsoft Grand Prize, Cartesia; general prizes as eligible.
- Backup demo video uploaded.
- Whole team at the assigned table from 10:30 AM to 12:30 PM Sunday [18].

14 Sources

[1] European Stroke Organisation Conference 2023: AI framework vs. call handlers on Copenhagen emergency calls. neuronewsinternational.com/ai-tool-outperforms-human-emergency-call-handlers-in-identifying-stroke/

[2] Wenstrup J., interview, ESOC 2024 (share of calls that were strokes). vjneurology.com/?p=8501

[3] Christensen H.C., interview, ESOC 2022 (accent and panic limitations; cardiac arrest system in Denmark). vjneurology.com/?p=5412

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> Medical note: LastWell is a hackathon prototype for decision support. Nothing in it is medical advice, and treatment decisions belong to the treating clinical team.